

```

syms t n;
Ta=4;
Tb=2;
f1=heaviside(t+1)-heaviside(t-1);
f2=sin(pi*t);

a0_a=(1/Ta)*int(f1,t,-Ta/2,Ta/2);
an_a=(2/Ta)*int(f1.*cos(2*pi*n*t/Ta),t,-Ta/2,Ta/2);
bn_a=(2/Ta)*int(f1.*sin(2*pi*n*t/Ta),t,-Ta/2,Ta/2);
disp(a0_a);
disp(an_a);
disp(bn_a);

a0_b=(1/Tb)*int(f2,t,0,Tb/2);
an_b=(2/Tb)*int(f2.*cos(2*pi*n*t/Ta),t,0,Tb/2);
bn_b=(2/Tb)*int(f2.*sin(2*pi*n*t/Ta),t,0,Tb/2);
disp(a0_b);
disp(an_b);
disp(bn_b);

```

(a)输出

$$1/2$$

$$(2*\sin((\pi*n)/2))/(n*\pi)$$

$$0$$

自上而下为a0, an, bn

$$\begin{aligned}
 a_0 &= \frac{1}{2} \\
 a_n &= \frac{2\sin\frac{n\pi}{2}}{n\pi} \\
 b_n &= 0
 \end{aligned}$$

(b)输出

$$1/\pi$$

$$\text{piecewise}(n == -2 \mid n == 2, 0, n \sim -2 \ \& \ n \sim 2, -(8*\cos((\pi*n)/4)^2)/(\pi*(n^2 - 4)))$$

$$\text{piecewise}(n == 2, 1/2, n == -2, -1/2, n \sim -2 \ \& \ n \sim 2, -(4*\sin((\pi*n)/2))/(\pi*(n^2 - 4)))$$

$$\begin{aligned}
 a_0 &= \frac{1}{\pi} \\
 a_n &= -\frac{8\cos^2\frac{n\pi}{4}}{\pi(n^2 - 4)} \\
 b_n &= -\frac{4\sin\frac{n\pi}{2}}{\pi(n^2 - 4)}
 \end{aligned}$$

```

syms t w
x1=exp(-1*t)*dirac(t-2);
x2=2*heaviside(2*t^2-4)-1;
x3=exp(-5*t)*heaviside(t+2);
x4=heaviside(t-1);
x1=fourier(x1, t, w);
x2=fourier(x2, t, w);
x3=fourier(x3, t, w);
x4=fourier(x4, t, w);
disp('傅里叶变换 x1 = ');
pretty(x1);
disp('傅里叶变换 x2 = ');
pretty(x2);
disp('傅里叶变换 x3 = ');
pretty(x3);
disp('傅里叶变换 x4 = ');
pretty(x4);

```

输出为

(1) 傅里叶变换 x1 =

$$\exp(-w \cdot 2i - 2i)$$

即 $e^{-2j(w+1)}$

傅里叶变换 x2 =
 (2)
$$\text{fourier}(\text{heaviside}(t^2 - 4), t, w)$$
 matlab不能处理阶跃函数中间加平方的傅

里叶变换。

手算结果为 $2\pi\delta(w) - \frac{4\sin\sqrt{2}w}{w}$

傅里叶变换 x3 =
 (3)
$$\frac{\exp(10 + w \cdot 2i)}{5 + w \cdot 1i}$$

即 $\frac{e^{10+2jw}}{5+jw}$

傅里叶变换 x4 =
 (4)
$$\exp(-w \cdot 1i) \cdot \pi \cdot \text{dirac}(w) - \frac{1i}{w}$$

即 $e^{-jw}(\pi\delta(w) + \frac{1}{jw})$